

1 Abstract

2 Climate change has shifted the timing of major life cycle events across most systems. These shifts
3 may in turn alter how species assemble temporally. While theory predicts that plant species should
4 partition time to reduce resource competition—for example by leafing out at different times—few
5 studies have examined this given the complexity of environmental cues that make timings highly
6 variable across years and sites. Here we examine species- and community-level responses in leafout
7 across 47 cuttings of woody plant species across four sites in North America. By manipulating known
8 temperature and photoperiod cues in controlled environments, then applying a phylogenetic Bayesian
9 model, we estimate the effects of these cues and evolutionary history in shaping leafout across species.
10 We found that each cue caused an advance in the day of leafout, but to varying degrees. Each of
11 these responses, however, were highly similar across sites, exhibiting instead greater variation across
12 species. Differences in species responses to well-studied temperature and photoperiod cues explained
13 only half of the observed variation in leafout date, with the remaining variation mostly explained by
14 the estimated phylogenetic structure of leafout and inherent species differences in the timing of their
15 growth. This suggests current models of leafout phenology are missing important variation that may
16 structure species assembly within communities. Identifying these unidentified cues or traits is critical
17 to forecasting phenological shifts and understanding their effects on community dynamics.

Introduction

Climate change has led to advances of about three days per decade in the timing of species life history events—phenology (Cohen et al., 2018; Parmesan, 2007; Thackeray et al., 2016). Different types of events, however, are highly variable, with certain events, such as leafout, spanning a period of weeks. Understanding and explaining this variability has become increasingly important as phenology shifts in response to rising temperatures at different rates (Fitter and Fitter, 2002; Fridley, 2012; Menzel et al., 2006; Parmesan, 2007; Yu et al., 2010). Some degree of variability is likely due to differences in climate change itself across space—as some areas warm faster than others and generate larger phenological shifts (Hoegh-Guldberg et al., 2018). Recent research suggests that species-level differences in phenology also drive some of this variation (Vitasse et al., 2009, 2018; Wolkovich and Cleland, 2014; Zohner and Renner, 2014). How these differences compare to other potential sources of variation, and what predicts them, is still largely unknown, but has major implications for predicting how changes in phenology will affect community dynamics and ecosystem services.

Differences in species phenologies can impact community structure and dynamics. Community ecology has long posited that species assemble in part through partitioning time, leading to unique temporal niches that allow species to coexist (Chesson et al., 2004; Grime, 1977). For example, earlier germination or leafout may provide understory species early-access to soil and light resources, before larger canopy species leafout (Heberling et al., 2019; Lee and Ibáñez, 2021). These different temporal niches may also trade-off different pressures, with research suggesting that in seasonal systems species that are active earlier experience greater abiotic pressures—such as spring frost—but avoid the greater competitive pressures for resources that occur later in the season when most species are fully active (Augspurger, 2009; Gotelli and Graves, 1996; Pau et al., 2011; Sakai and Larcher, 1987).

As climate change variably affects the timing of growth, it may alter the temporal order of species in a community and result in novel interactions between species (Cleland et al., 2012; Rudolf, 2019; Tiusanen et al., 2020), thus changing the competitive landscape and synchrony of interactions within communities. How consistent these re-orderings are would depend then on the climatic changes in any one location, the species composition of the community and potentially any local adaptation of phenological cues that may drive population-level differences within species. The relative importance of local adaptation of phenological cues is currently debated, with some research suggesting how plastic or locally adapted phenological cues are depends on the species and phenophases (Inouye et al., 2010; ?).

While recent changes in climate are shaping differences in phenology, for many species their temporal

52 niches may also reflect their evolutionary history. Species traits in communities today are the result of
53 selective pressures acting over evolutionary timescales that include the historical climates and related
54 ancestral phenotypes that were favourable under previous conditions (Wiens et al., 2010). This could
55 produce greater similarity in phenological traits of more closely related species (Davies et al., 2013;
56 Morales-Castilla et al., 2024), which would impact how species respond to future climates.

57
58 Spring budburst offers an excellent system to test for species- and community-level patterns in phe-
59 nology and environmental cues. Budburst of temperate woody plants is well studied and thus known
60 to respond to environmental cues, especially temperatures in the winter and spring—referred to as
61 chilling and forcing, respectively—and daylength or photoperiod (Basler and Körner, 2014; Chuine
62 et al., 2010; Cooke et al., 2012; Laube et al., 2014; Polgar and Primack, 2011). Budburst responses
63 to these cues are generally similar across sites and populations (Aitken and Bemmels, 2016; Zeng and
64 Wolkovich, 2024). Species-level differences, however, are generally much larger than observations of
65 site or population-level variation (but see Deans, 1996; Sogaard et al., 2008, for examples of population-
66 level differences). Decades of phenological research suggests these unique phenologies are likely due
67 to different responses to temperature and photoperiod cues, though they may be also structured by
68 species shared evolutionary history (Davies et al., 2013; Gougherty and Gougherty, 2018; Lechowicz,
69 1984) and their current environments. Forest understory species, for example, should vary their leafout
70 and budburst early to both gain access to light and soil nutrients prior to canopy closure (Mahall and
71 Bormann, 1978; Muller, 1978), and to reduce competition between species within the understory versus
72 canopy.

73 Here we test how environmental—temperature and photoperiod—cues and evolutionary relationships
74 shape species-level variation in phenology of forest communities across North America. We leveraged
75 two large-scale growth chamber cutting experiments, in which we vary the temperature cues of chilling
76 (cool temperatures in the fall and winter) and forcing (warmer temperatures, usually in the spring) with
77 photoperiod (Fig. 1), with a Bayesian phylogenetic approach to estimate budburst responses across 47
78 woody species. We collected samples (cuttings) from adult plants at four sites, in eastern and western
79 North America (Fig. 1), with pairs of sites on each coast spanning 4 and 6° latitude, respectively.
80 Our dataset includes the diverse assemblages of tree and shrub species found in eastern and western
81 deciduous forests, allowing us to explore differences across both sites and functional groups—including
82 understory shrubs versus canopy trees—in eastern versus western forests.

Materials and Methods

Field sampling

We combined data from two growth chamber studies using dormant branch cuttings of North American deciduous woody plants. Such cutting experiments are a common and powerful approach to infer the responses of adult trees to environmental conditions (Primack et al., 2007; Vitasse et al., 2014). In our first study, we collected samples from two eastern communities—Harvard Forest, Massachusetts, USA (42.55°N, 72.20°W) and St. Hippolyte, Quebec, Canada (45.98°N, 74.01°W), from 20-28 January, 2015 (previously reported in Flynn and Wolkovich, 2018). The second growth chamber study spanned two western communities—Smithers (54.78°N, 127.17°W) and E.C. Manning Park (49.06°N, 120.78°W), British Columbia, Canada, with sampling from 19-28 October 2019 (not previously published). Combining the datasets from these two experiments allowed us to test for spatial variation in the phenology of woody plant communities at a larger geographic scale and make stronger inferences for how different species vary in their cues.

We selected the dominant deciduous species in each forest community and maximized the number of species occurring across sites. We observed budburst for 47 species spanning eastern and western forest communities, with 28 species at our eastern transect and 21 species at our western transect, of which 3 species occurred at both regions (Fig. 1, Table S1). Of the species in our eastern transect, there were 13 shrubs and 15 trees, while our western transect included 18 shrub and 4 tree species. At all sites, we tagged 15-20 healthy, individuals for each species in the summer prior to sampling. To reduce ontogenetic effects (Vitasse et al., 2014), we only sampled mature individuals (e.g. for trees we sampled individuals greater than 2 m in height and with DBH greater than 4 cm, Natural Resources Canada, 2020, and only sampled shrubs with woody stems). We collected a single cutting from small shrub species, but depending on the size of the sampled tree, we took between 6-16 cuttings, thus collecting samples from 6-20 tagged plants per species. All cuttings were taken from the same relative canopy position, sampling terminal branches from shrubs and the lower canopy from trees using a pole pruner from the ground. We kept samples cold and immediately placed them in erlenmeyer flasks of water upon returning from the field. We conducted the eastern study at the Arnold Arboretum, Boston, MA, USA, and the western study at the University of British Columbia, BC, Canada. We used daily weather data from local weather stations to calculate the field chilling prior to collection (see Table S2).

Growth chamber study

We used growth chambers with two levels of chilling—with either no additional chilling or 30 days at 4°C for our eastern study, and 21 or 56 days of chilling at 4°C for our western study (all dark)—after which we moved plants to one of two levels of forcing—with day and night temperatures that varied to create a cool regime of 15:5°C and a warm regime of 20:10°C—crossed with two photoperiods of either 8 or 12 hours—for a total of 8 distinct treatments in each study.

Our design was similar for both eastern and western sites (Fig. 1). We had seven to ten replicates per treatment in our eastern study and eight replicates in our western study, with the exception of *Sorbus scopulina*, for which we only had six replicates per treatment. Our eastern and western studies differed only in the timing of sample collection and thermoperiodicity (the interval of alternating day and night temperatures) in forcing treatments. By collecting samples in our eastern study in late January, they experienced considerable field chilling. We collected the western samples in late October, with most chilling occurring under controlled conditions. In our eastern study, we also set the duration of forcing temperatures to correspond with the photoperiod treatments. While this is a widespread and common approach in phenology chamber studies, it leads to a confounding effect in the thermoperiodicity (Buonaiuto et al., 2023). Thus we applied forcing temperatures for 12 hours independent of the duration of the photoperiod treatments in our western study, and controlled for this difference in our statistical approach (see below). Each experiment was conducted across 8 growth chambers. We additionally rotated cuttings within chambers approximately every two weeks to avoid any chamber effects and changed the water and re-cut each cutting to prevent callusing and keep cuttings healthy. We controlled for small differences in the number of replicates across species in our statistical approach using partial pooling (see section on statistical analysis).

We assessed phenological stages of budburst using the BBCH scale (Finn et al., 2007), adapted for our specific species. We also created photographic guides to help define each stage of the BBCH scale for our specific species (Loughnan and Wolkovich, 2024; Savas et al., 2017). Every 1-3 days, we observed each sample and recorded all phenological stages up to full leaf expansion (defined as code 17 by Finn et al., 2007). Here, we modelled budburst timing, defined as the day of budbreak or shoot elongation (denoted as code 07 by Finn et al., 2007) since forcing treatments began. Our eastern study spanned 82 days, with over 19320 phenological observations, while the western study spanned 113 days and 47844 phenological observations.

Statistical Analysis

We tested for differences in budburst cues across species and communities using a phylogenetic mixed effects model with partial pooling ('shrinkage') across species. This approach accounts for both the evolutionary relatedness of our species and estimates the species-level cues and differences across sites. We obtained species phylogenetic relatedness by pruning the Smith and Brown (2018) megatree of angiosperms (Fig. 2).

In our model, we included each of our three cues as continuous variables. To account for the later collection dates and greater amounts of field chilling experienced in our eastern study, we converted our chilling treatments to chill portions for each individual community using local weather station data (Table S2) and the chillR package (v. 0.73.1, Luedeling, 2020). Chill portions are a commonly used chill metric that is calculated using hourly climate data and a Dynamic Model in which cool temperatures lead to the accumulation of chilling, while more accurately estimating chilling accumulation under fluctuating temperatures (Luedeling, 2012, 2020). We calculated chill portions for both the chilling experienced in the field prior to sampling and during the experiment in our 4°C chilling chambers. We also converted forcing temperatures to mean daily temperatures in each treatment to account for differences in thermoperiodicity between the two studies (Buonaiuto et al., 2023). Finally, we z -scored each cue and site using two standard deviations to allow direct comparisons between results across cues (Gelman, 2008).

Our modelling approach allowed us to combine observations of budburst (y_i) across species (sp) to estimate both the effects of species-specific differences in their responses (the rate of change in budburst day per unit of standardized cue, as estimated by the slope, β) to cues (chilling, forcing and photoperiod). By z -scoring the model coefficients, the estimates of the species-level intercepts (α_{sp_i}) represent species differences relative to the mean day of budburst (Gelman et al., 2020). Our full model also estimated effects of sites and all possible two-way interactions between cues, and between cues and sites, to estimate the day of budburst ($\hat{y}$) relative to the first day of forcing conditions and

relative to baseline conditions and our our most northern site (Smithers).

$$\begin{aligned}\hat{y}_i = & \alpha_{sp_i} + \beta_{Manning\ park} + \beta_{Harvard\ forest} + \beta_{St.\ Hippolyte} + \\ & \beta_{chilling_{sp_i}} + \beta_{forcing_{sp_i}} + \beta_{photoperiod_{sp_i}} + \\ & \beta_{forcing \times chilling_{sp_i}} + \beta_{forcing \times photoperiod_{sp_i}} + \beta_{chilling \times photoperiod_{sp_i}} + \\ & \beta_{forcing \times Manning\ park_{sp_i}} + \beta_{forcing \times Harvard\ forest_{sp_i}} + \beta_{forcing \times St.\ Hippolyte_{sp_i}} + \\ & \beta_{chilling \times Manning\ park_{sp_i}} + \beta_{chilling \times Harvard\ forest_{sp_i}} + \beta_{chilling \times St.\ Hippolyte_{sp_i}} + \\ & \beta_{photoperiod \times Manning\ park_{sp_i}} + \beta_{photoperiod \times Harvard\ forest_{sp_i}} + \beta_{photoperiod \times St.\ Hippolyte_{sp_i}}\end{aligned}$$

$$y_i \sim \text{normal}(\hat{y}_i, \sigma^2)$$

The slopes were modelled with partial pooling at the species-level:

$$\begin{aligned}\beta_{forcing_{sp}} & \sim \text{normal}(\mu_{forcing_{sp}}, \sigma_{forcing}^2) \\ \dots & \\ \beta_{photoperiod \times St.Hippolyte_{sp}} & \sim \text{normal}(\mu_{photoperiod \times St.Hippolyte_{sp}}, \sigma_{photoperiod \times St.Hippolyte_{sp}}^2)\end{aligned}$$

We included the phylogenetic effect on species-specific differences in budburst timing (α_{sp}) as a variance covariance matrix ($\mathbf{V}$) in the parameterization of the normal random vector:

$$\boldsymbol{\alpha}_{sp} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \dots \\ \alpha_n \end{bmatrix} \text{ such that } \boldsymbol{\alpha} \sim \text{multi-normal}(\mu_{\alpha}, \mathbf{V})$$

The timing of budburst ancestrally is represented as the root value of the means of the multivariate normal distribution. The phylogenetic structuring of budburst is estimated using Pagel's lambda, a phylogenetic scaling parameter where values close to one indicate a phylogenetic signal consistent with Brownian motion and values close to zero the absence of a phylogenetic signal for a given trait (Pagel, 1999; Pearse et al., 2025). For more detail on this method of phylogenetic modeling, see Morales-Castilla et al. (2024). In addition to our main model, we performed a subsequent analysis to check our phylogenetic results were robust to whether we included the phylogenetic effect on the intercept (shown above) or on the residuals instead of the intercept, which is more similar to the common approach of PGLMM (Pagel, 1999; Pearse et al., 2025); for this we used the same model framework but

189 moved the phylogenetic effect from the intercept to σ_y^2 .

190
191 We used non-informative priors for each of our model parameters (increasing priors three-fold did not
192 demonstrably change our estimates), and validated the model code using test data prior to analysis.
193 Our model produced $\hat{R}$ values close to 1, well mixed chains, and n_{eff} values that exceeded 10% of the
194 model iterations. Our model was fit using the Stan language with the rstan package (v. 2.26.23, Stan
195 Development Team, 2018) in R (v. 4.3.1, R Development Core Team, 2017).

196
197 We present model estimates, relative to baseline or low treatment conditions. This means that we
198 show the average daily temperature for our cooler forcing conditions (which averaged to 8.33°C for our
199 eastern sites and 10°C for our western sites, after accounting for the differences in thermoperiodicity),
200 with a short photoperiod (8 hours photoperiod), and the mean of lowest chill portions from each pop-
201 ulation (which is equivalent to approximately 10 weeks of chilling at 4°C) and—when relevant—our
202 most northern site (Smithers). We present results as means and 90% posterior uncertainty intervals
203 from our Bayesian phylogenetic models.

205 Results

206 On average, species budburst occurred 30.4 days (uncertainty interval: 26.0, 35.2) after the start of
207 forcing and photoperiod treatments (all estimates are given as mean $\pm$ 90% uncertainty intervals,
208 henceforth ‘UI,’ and budburst dates given as relative to baseline conditions, see methods for more de-
209 tails). Between the earliest and latest budbursting species, the difference in the mean day of budburst
210 was 38.5 and 30.7 days for the eastern and western sites respectively. Budburst of individual species
211 showed distinct differences in their timing and the relative importance of cues, but all cues led to an
212 advance in budburst date. Of the three cues, chilling had the largest effect, with increased chilling
213 advancing budburst (we represent advances with negative numbers, while delays are positive; -14.8,
214 UI: -16.7, -12.9 days per standardized chill portions) and photoperiod the smallest (-3.4, UI: -4.0,
215 -2.7 days per standardized photoperiod, Fig. 3a, S1). But we found a large, subadditive interaction
216 between forcing and chilling (8.4, UI: 7.0, 9.8, Fig. 3b); meaning that low chilling is offset by high
217 forcing conditions, and vice versa (Table S3).

218
219 Overall we found species budburst was moderately phylogenetically structured (Pagel’s λ of 0.4, UI:
220 0.1, 0.8, Fig 2, estimated on the intercept; results were similar when estimated on the residuals XXXX).
221 This phylogenetic structure suggests latent processes that shape species-level differences in budburst

222 timing beyond the species-specific differences in their shifts responses to temperature (chilling and
 223 forcing) and photoperiod. In modelling the phylogenetic structure on the residuals, estimates of
 224 phylogenetic structure were similar (λ : 0.5, UI: 0.1, 0.9; Table S4), providing further support that the
 225 species-level variation not explained by the cues is in part attributable to evolutionary history. The
 226 contribution of these species-specific cue differences to budburst order and timing was small compared
 227 to the overall species differences unexplained by cues (represented in the modeling framework as the
 228 intercept), with the exact magnitude for which chilling, forcing and photoperiod determine budburst
 229 date depending on the level of each cue (e.g., forcing has a much larger effect on budburst timing given
 230 a constant daytime temperature of 20°C compared to a constant daytime temperature of 15°C). Across
 231 the levels of chilling, forcing and photoperiod we studied, species-specific differences in responses to
 232 cues explained between 37.6% and 67.8% of variation for eastern species and between 47.4 and 61.7%
 233 for western species, with the phylogenetically structured intercept explaining the rest (Fig. 2).

234
 235 We also found the rank order of species' budburst timings changed more when estimated using higher
 236 chilling (which is equivalent to approximately 15 weeks chilling at 4°C when we account for the aver-
 237 age field and experimental chilling plants experienced across our two experiments), forcing and longer
 238 photoperiods than under lower chilling and forcing and shorter photoperiods (Fig. S2). The budburst
 239 ranking for the three species that occurred in both transects remained relatively consistent across our
 240 eastern and western sites (Fig. 4), with only *Alnus incana* in the western sites experiencing a large
 241 change in rank with cues (Fig. S2). In comparing the earliest and latest budbursting species, we
 242 found relatively small differences in cues (Fig. 4). For example, an early budbursting shrub, *Lyonia*
 243 *ligustrina*, had chilling and forcing cue estimates of -14.7 and -11.5 respectively, which are comparable
 244 to the cue estimates of -15.7 and -11.5 for the much later tree, *Quercus alba*. Yet the model predicts
 245 these species to have a 15.1 day difference in day of budburst, highlighting how the timing of budburst
 246 across species was also strongly related to species level differences (intercept values) independent of
 247 their responses to chilling, forcing and photoperiod (see Fig. 5).

248
 249 Despite our expectations that species responses should differ across the canopy layer, with shrubs
 250 being less sensitive to cues than trees, we did not find substantial differences across these two func-
 251 tional groups (Fig. S3). Shrubs, like *Cornus stolonifera*, budburst early and showed small responses
 252 to chilling and forcing (Fig. 4b, d). But 36.7% of the shrubs, including *Menziesia ferruginea* and
 253 *Symphoricarpos alba*, exhibited the opposite response and budburst relatively late (Fig. 5, S3). Simi-
 254 larly for trees, some species did match our predictions, exhibiting stronger cues, but 23.5% of species
 255 budburst earlier than expected. While specific tree species, such as *Quercus velutina*, did have larger
 256 chilling and photoperiod responses as predicted (Fig. 4a, e), and *Fagus grandifolia* had the largest

photoperiod response (Fig. 4e), overall we did not find clear differences between the cues of trees and shrubs across the four sites.

Finally, we observed negligible site-level effects compared to differences between cues (Fig. 3, Table S3). Treatments that received a month less of chilling (low chilling), that were 5°C cooler (low forcing) with shorter photoperiods, all budburst later compared to the more chilled, warmer and long photoperiod treatments across all sites and across transects (Fig. S1 a-c, shown for baseline conditions, see Table S3 for model output). Overall budburst dates did not differ between sites, though eastern sites budburst marginally earlier (day 35.8, UI: 28.3, 42.3) compared to the western sites (day 48.1, UI: 40.7, 54.5, see also Table S3 for model output). This could be due to the earlier collection date of our western species, which reduced the field chilling they received relative to our eastern samples (though see corrections for this in the statistical analysis section of our methods). The substantial overlap in budburst dates overall, and when estimated with low and high cues (see Fig. S1 supports that the forest communities at these different sites are highly similar, despite the large distances and species compositions between them.

Discussion

To accurately forecast shifts in phenology and its impact on ecological communities requires understanding which environmental cues are most important at the population-, species-, functional-group and community-levels. Our results support decades of research finding that budburst timing advances in response to the well-studied major temperature cues of chilling and forcing (with the greatest responses to chilling and forcing respectively, see also Ettinger et al., 2020; Flynn and Wolkovich, 2018) and longer photoperiods. But using our modelling approach allowed us to estimate the extent to which these species-specific responses to these cues do—or do not—lead to differences in species timings that may structure temporal niches within communities. Perhaps surprisingly, we found a large portion (one-third to one-half) of variation in species budburst timings was unexplained by different responses to chilling, forcing and photoperiod. Our results show that some of this variation is due to species evolutionary history, but not all, suggesting fundamental gaps in our model of one of the best studied and most important phenological events—woody plant budburst.

Differences in species responses to temperature and photoperiod were similar across sites and functional groups, suggesting that understanding the drivers of variation across species may be an important task to improve forecasting. Our study included 47 of the dominant woody species from deciduous forests

across North America from both shrub and tree functional groups, with a mix of early and late budbursting phenologies. Despite the high degree of diversity and spatial scale represented in our study, we found no detectable differences in the responses to temperature and photoperiod at our different sites, even though sites span different coasts and over 6° of latitude. This contrasts with research that suggests latitudinal trends in photoperiod and temperature may impose unique selective pressures, and thus may drive site-level differences in responses to cues (Keller et al., 2011). Differences in spring phenology have been found across latitudinal by studies using *in situ* phenological data (Alecrim et al., 2023; Post et al., 2018), as well as elevation gradients based on common garden studies, with different studies attributing this variation to either genetic effects (Gugger et al., 2015; Vitasse et al., 2009) or differences in local microclimates (McDonough MacKenzie et al., 2018; Stinston, 2005; Vitasse et al., 2013). Many of these studies, however, are based on long-term observations in the field, with confounding differences in study duration and start dates, variable methodologies, and geographic extent. The lack of site-level effects that we observed—using experiments to estimate responses—suggests budburst cues are not shaped by local conditions, at least at our study scale.

Community composition and interspecific variation in phenology

The variation across species in their responses to environmental cues has the potential to create large differences in species temporal niches and ecological roles. Species ranged from early to late budburst timing in our experimental conditions, spanning a similar period as observed in natural communities (Lechowicz, 1984; Richardson, A.D., O’Keefe, 2009). This suggests our experiment captured a realistic breadth in temperate forest budburst. Furthermore, all our focal species responded to each environmental cue, with large responses to chilling and comparatively small responses to photoperiod—trends consistent with previous studies (Ettinger et al., 2020; Flynn and Wolkovich, 2018). Cues, however, had complex interactions that appear potentially advantageous under warming climates (Caffarra and Donnelly, 2011; Flynn and Wolkovich, 2018; Heide, 1993). The interaction between chilling and forcing may ensure that species budburst even if warmer winters cause insufficient chilling, but will require additional forcing. But despite these differences in species responses to cues, we did not find clear, generalizable trends across functional groups.

Shrubs and trees differ greatly in their physiology, filling different ecological niches. Most of our trees budburst later than our study’s shrubs, a relative order also found by previous studies (Gill et al., 1998; Panchen et al., 2014; Polgar et al., 2014). But about a quarter of our trees also budburst early, with timing more similar to that of shrubs. Similarly, we found a third of shrubs budburst at similar

times as the majority of our trees. These deviations from our expectations for how shrubs and trees partition their budburst timing suggests there is more nuance to these patterns. As tree species advance phenologically—resulting in earlier canopy closure and reduced light availability (Donnelly and Yu, 2019; Heberling et al., 2019)—this subset of shrubs with small cues may have reduced fitness. Our findings suggest, however, that many shrubs will also advance phenologically and have the potential to maintain their relative temporal niche.

Community assembly in responses to cues versus evolutionary history

Our study included the three cues—chilling, forcing and photoperiod—commonly thought to determine budburst timing, but over a third of the total variation was not explained by these cues (Fig. 4), suggesting the general model of budburst may be incomplete. This, paired with the observed phylogenetic structure in budburst timing, hints at unidentified latent traits still missing from our understanding of the processes shaping spring phenology (Davies et al., 2019; Webb et al., 2002).

Latent traits could reflect other cue or trait relationships indicative of species growth strategies or key traits that may facilitate greater resource use, or the greater competitive abilities needed when budbursting later (Grime, 1977; Inouye et al., 2010). For example, species that budburst later in the season may have more conservative growth strategies, possessing traits that allow them to compete for limited soil resources and light, such as low specific leaf areas or higher wood densities (Chave et al., 2009; Wright et al., 2004). But our results suggest that evolutionary history and species ancestral phenotype also have some effect on species timings and cues. Species traits are shaped over ‘deep’ timescales, with previous evolution also influencing responses to the present climate. Phenological trends could thus be an indication of limited evolutionary time to adapt (Lechowicz, 1984), or shaped by selection on other important traits that are not phylogenetically structured. Incorporating both phenology and evolutionary history into a broader trait framework could thus provide insights into traits that correlate with budburst timing, and—ultimately—the current and evolutionary drivers that select for species phenotypes and possibly shape species adaptive potential to future climates.

Predicting budburst under future climates

As climate change leads to greater temperature cues, we are already observing advances in species budburst within diverse ecological communities. These changes are likely to have cascading impacts on species interactions, with greater overlap in growth timing potentially increasing competition for

pollinators, resources, or light. Over time, such impacts could impact species performance and alter species assemblages. The greatest ecological impacts are likely to occur in high latitude communities that are warming the most and the fastest (Hoegh-Guldberg et al., 2018). Our results on the effects of cues on rank order suggest this extreme warming could alter the order with which species budburst, further creating novel species interactions relative to communities experiencing smaller changes in cues. Yet our ability to predict these changes is often limited by a lack of available data for most species and biases in which species have been studied to date.

In addition to our findings providing insights into the phenological cues of understudied temperate forests communities, our approach encompasses several recent advances in forecasting phenological responses to climate change. By accounting for shared evolutionary history when estimating species-level variability, we were able to better understand cue responses for poorly sampled species and infer responses for closely related species for which data is unavailable (Molina-Venegas et al., 2018). We can therefore leverage our existing data to still develop meaningful forecasts of future community change and develop more effect pilot studies for species that are rare, protected, or challenging to study. Secondly, the consistency of cue responses we observed across communities suggests that species responses to future climate may be consistent at a regional spatial scale. As such, our understanding of forest community responses to climate change may benefit more from studies that encompass high taxonomic diversity and thus species-level variation, rather than the study of a few species over larger spatial scales.

While our findings have applications to plant communities, our analytical approach is widely applicable to forecasting diverse species responses to climate change. The phylogenetic model we used here could be easily applied to other phenological events or suites of species from across the tree of life. By combining species-rich and community-wide approaches with phylogenetic relationships we can better understand the impacts of evolutionary history and current ecological processes in shaping species phenology under a changing climate.

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390 **Author Contribution Statement**

391 Both D.L. and E.M.W. conceived the study, contributed to the analysis and code and contributed to
392 the writing and revision of the manuscript. DL led field and experimental data collection for western
393 sites.

394

395 **Competing Interests Statement**

396 The authors have no competing interests.

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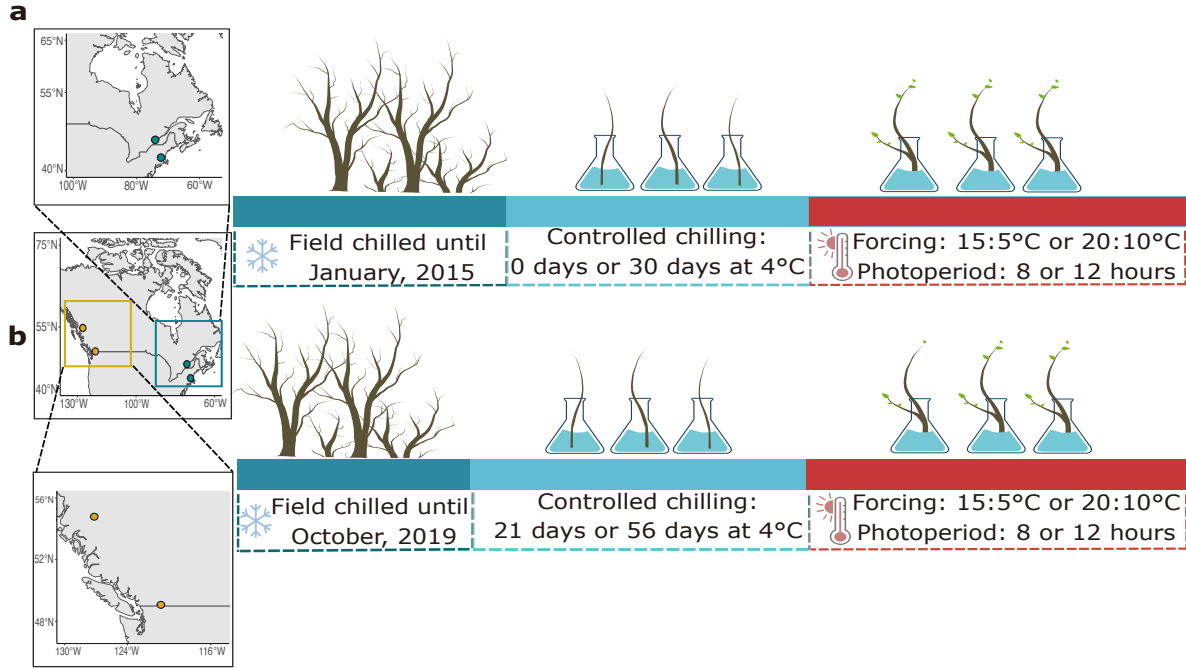
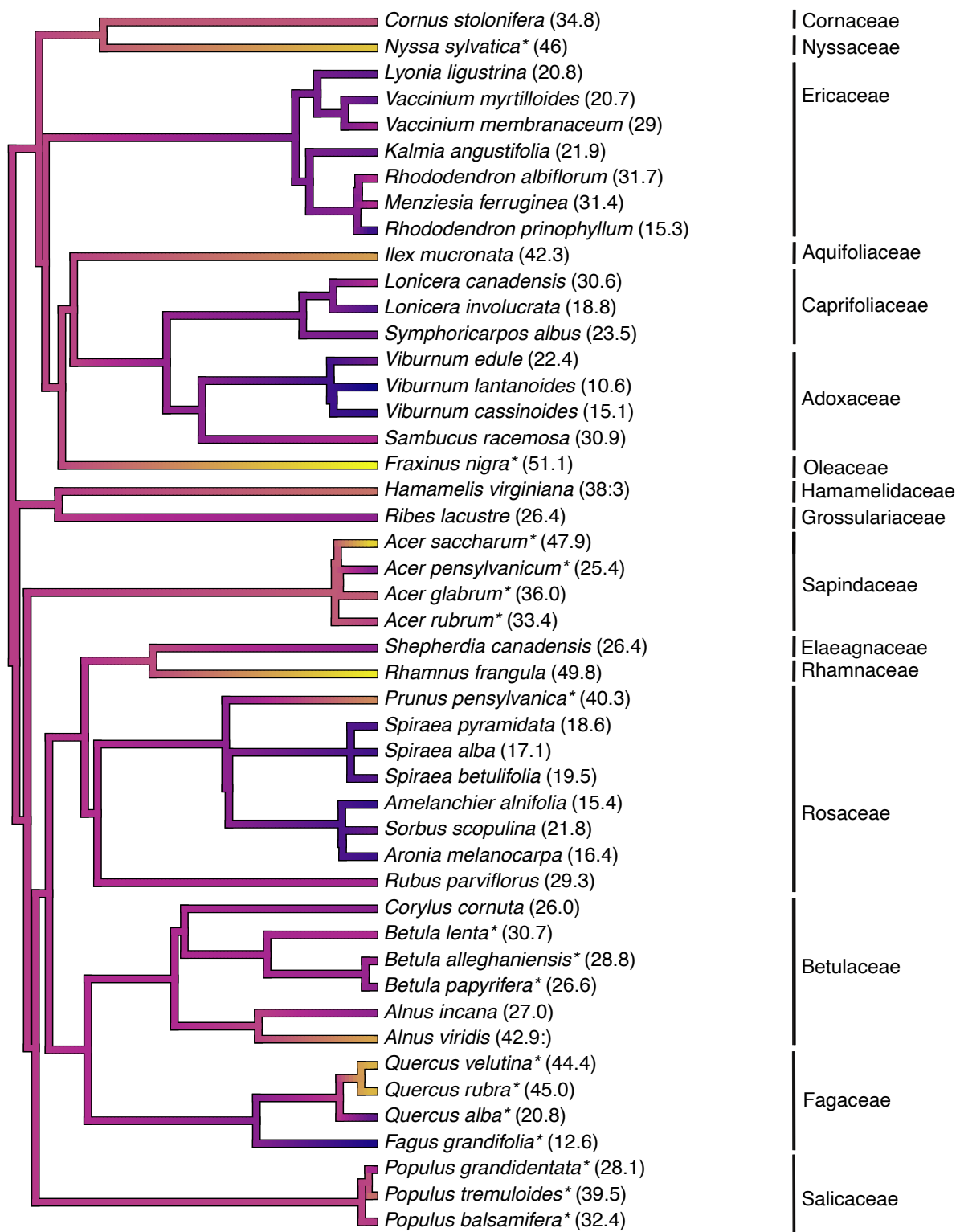


Figure 1: We combined data from two controlled environment studies of temperature and photoperiod cues for budburst in temperate woody species **a**, from two forest communities in eastern North America that span a 4° latitudinal gradient and **b**, two communities in western North America that span a 6° latitudinal gradient. Combined, our dataset includes 28 tree and shrub species from eastern communities and 21 species from western communities, with three species occurring across all four populations. Both studies used a full-factorial design with two levels each of chilling treatments (consisting of a combination of field and different duration of chamber chilling), forcing and photoperiod, for a total of eight unique treatments. Observations of budburst were made during the forcing treatment, using the BBCH scale to classify stages of bud development (see Loughnan and Wolkovich, 2024; Savas et al., 2017, for guides we made for these species).



10 Intercept (days) 50

Figure 2: Tree (denoted with an asterisk) and shrub species differences were accounted for by including phylogenetic effects on the intercept of a model estimating days to budburst after the start of forcing treatments (simplified to 'days₂₃ here'). This approach estimated a phylogenetic scaling parameter for the intercept of 0.4 (where values close to one indicate a phylogenetic effect consistent with Brownian motion and values close to zero the absence such an effect). Included for each tip of the phylogeny is the species-level intercept value in brackets, along with the names of major the clades represented in the tree.

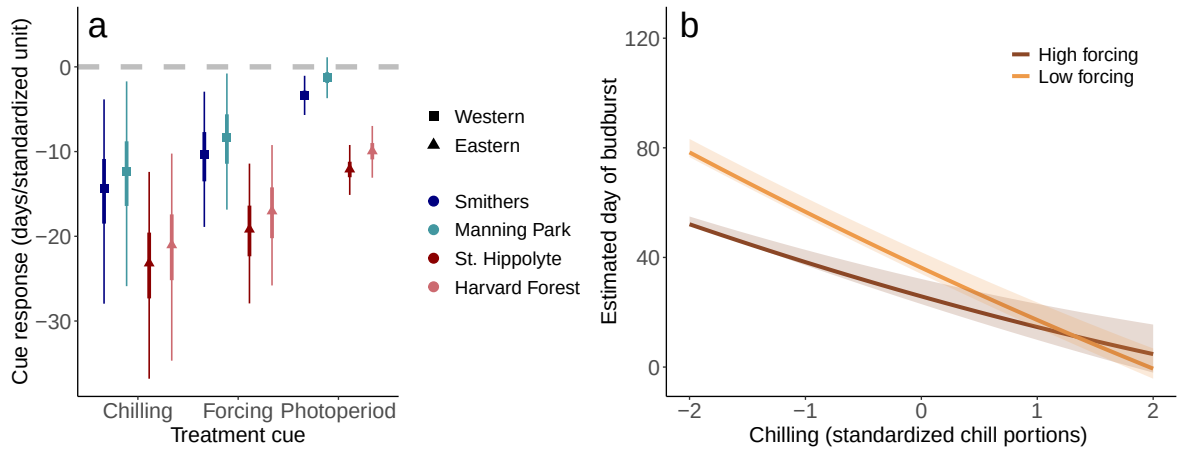


Figure 3: **a**, Posterior distributions of estimated changes in budburst per unit change in chilling, forcing, and photoperiod with individual site-level effects. Symbols denote the eastern (triangle) and western (square) transects and the darker point color our more northern communities. Points represent the mean, thicker lines the 50% uncertainty interval, and the thin lines the 90% uncertainty interval. **b**, Cues interacted to produce a subadditive effect, where delays in budburst under low chilling was offset by high forcing conditions, and vice versa. To make results comparable across cues we standardized chilling, forcing, and photoperiod via z -scores using two standard deviations, see methods for further details.

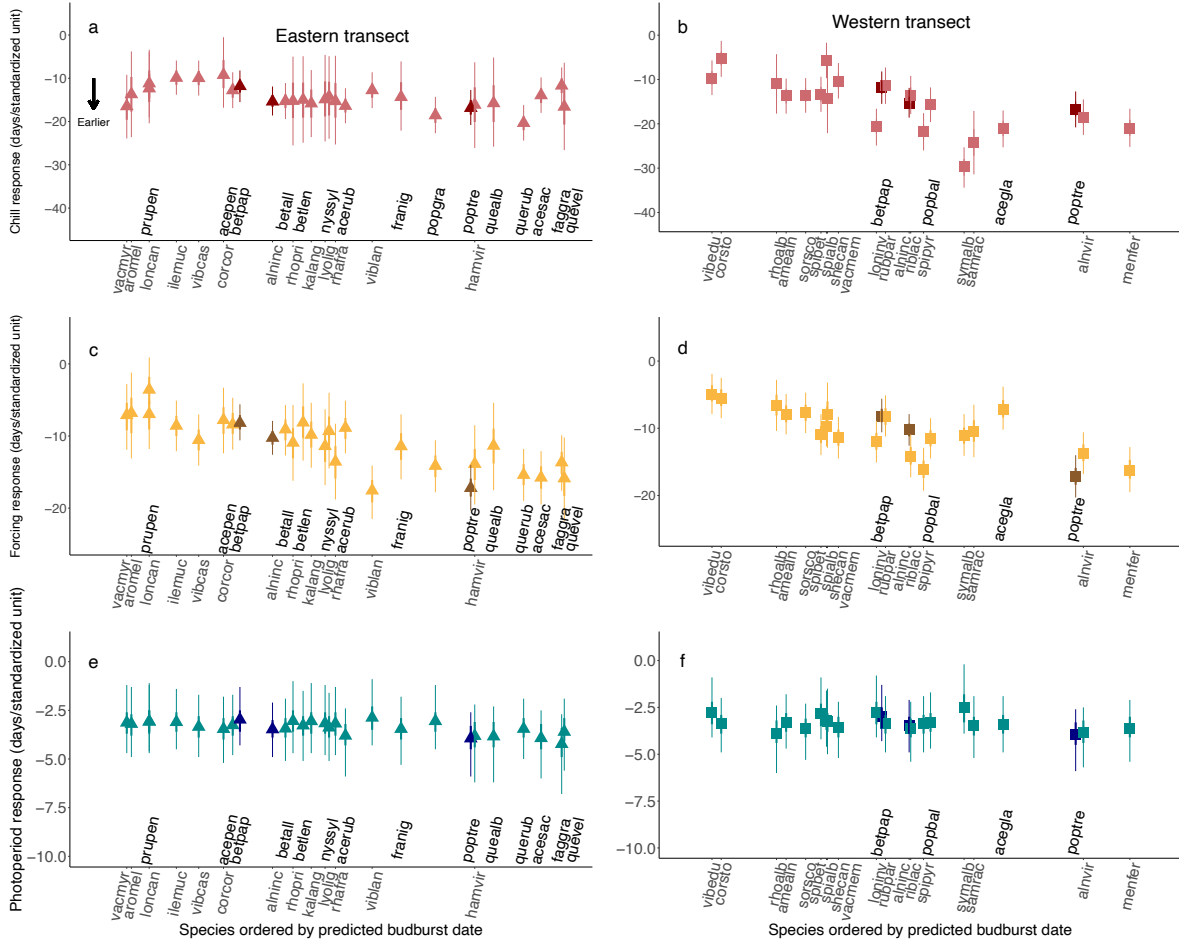


Figure 4: **Estimated change in species budburst day per unit of standardized a,b chilling, c,d forcing, and e,f photoperiod**, ordered by estimated budburst dates for both the **a,c,e**, eastern and **b,d,f** western sites based on our model estimates. Cues are plotted on differing y-axis scales to better depict species differences across cues. For each species, the points represent the mean, thinner error bars represent the 90% uncertainty interval, and thicker bars represent the 50% uncertainty interval. Points for the three species that occur in both the eastern and western transects are depicted using darker colours. **Tree species are labeled above the x-axis and shrub species below the x-axis**. To make the results across cues directly comparable, we standardized predictors (here, chilling, forcing, and photoperiod) via z -scores using two standard deviations, see methods for further details.

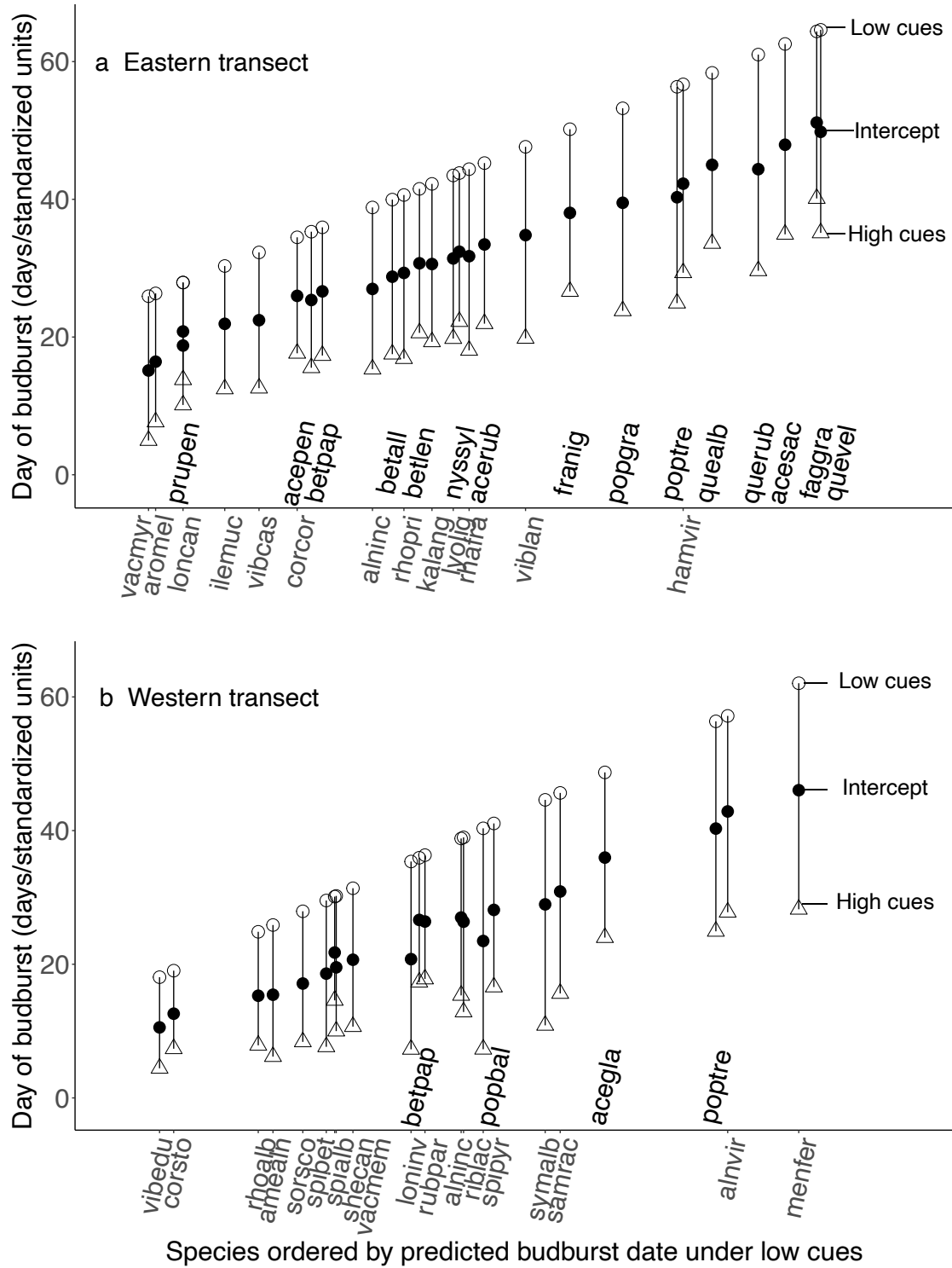


Figure 5: Comparisons of estimated day of budburst for tree (labeled above the x -axis) and shrub species (labeled below the x -axis) based on the full model (intercept plus all cues), shown as white (open) symbols, versus the intercepts only (without effects of chilling, forcing, and photoperiod), shown as black (filled) symbols. Points are ordered according to differences in species estimated budburst dates for both the **a**, eastern **b**, and western sites, with low cue conditions, depicted as circles, and high cue conditions, depicted as triangles.